AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY (AUST)

Department of Computer Science and Engineering

Final Project Report

Course No: CSE3224
Course Title: Information System Design & Software Engineering Lab

Project Title: **"The Parody Journal: A Broad-Spectrum AI-Enhanced Interactive News Platform"**

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Submitted to

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Report on The Parody Journal Project

1. Reason for Project Selection With information surrounding us in today's world, readers find it difficult to find reliable, pertinent, and engaging news. Most of the platforms available either cover generic news or niches, lacking in offering a well-balanced mix of personalized content on various topics like technology, lifestyle, entertainment, politics, and global issues. Further, the absence of interactive features and engaging multimedia makes the readers feel alone and inactive.

2. Proposed Solution The Parody Journal will be an interactive news website that will:

- **Diversify Content:** Cover a wide range of topics, including technology, politics, entertainment, lifestyle, health, and global trends.
- **Integrate AI:** Offer personalized recommendations, content briefs, and sentiment analysis.
- **Incorporate Multimedia:** Mix written articles with videos, podcasts, and infographics to offer richer storytelling.
- **Foster Engagement:** Provide interactive elements like commenting, surveys, and live Q&A sessions with specialists.

3. Goals

- Create a unified and diverse consumption platform for news.
- Leverage AI to improve content discoverability and personalization.
- Provide an immersive multimedia experience.
- Create an easy-to-use .NET-based platform with scalability and performance.

4. Background Study

1. **Diverse Content Demand:** Research shows a high user preference towards sites that mix general and special content.
2. **Multimedia Integration Trends:** Video, podcasting, and interactive graphics are essential to success with digital storytelling.
3. **Use of AI in News:** AI-driven features greatly enhance user experience through personalized recommendations and content analysis.
4. **Effect of Interactive Features:** Community interaction features lead to increased user engagement and platform loyalty.

5. Requirement Analysis

Functional Requirements

Designing a User: A user should be able to:

- Create and manage their profile with personal details.
- Personalize their news feed preferences based on their preference.
- Track their history of reading.
- Bookmark articles for later reading.
- Personalize account settings and privacy controls.

User Authentication and Management: Users should be able to:

- Sign up via their email.
- Log in securely using a secure password.
- Reset passwords via email verification.
- Modify their profile information.
- Delete their account with all associated data.

Content Personalization: Users should be able to:

- Choose their preferred news categories.
- Follow specific topics or tags.
- Rate articles and give their feedback.
- Save their reading behavior.

News Feed Management: The system will:

- Show personalized news based on user preferences.
- Organize content by relevance, date, or popularity.
- Filter content by category, source, or topic.
- Refresh content in real-time.

Social Features: Users should be able to:

- Share articles on their social networks.
- Comment on articles and have discussions.
- Create and participate in topic-based communities.
- Take part in polls and surveys.
- Take part in live Q&A sessions.

Notification System: Users will be alerted on:

- Breaking stories from selected categories.

- Latest updates on topics they are tracking.
- Responses to comments.
- Live update alerts.
- Content suggestions based on the user's individual content requirements.

AI-Augmented Features: The system will give:

- Summaries of lengthy articles.
- Sentiment analysis of comments and trends.
- Personalized reading suggestions based on individual user requirements.
- Tagging and categorizing of content.
- Translation service for global news.
- Trend analysis with predictions.

Non-Functional Requirements

Performance:

- Page loading needs to be less than 3 seconds.
- Support for a minimum of 10,000 concurrent users.
- API call response time should be under 200ms.
- Smooth scrolling and navigation.
- Efficient content caching.
- Optimized image and video loading.

Security:

- We employ secure authentication methods.
- Regular security audits.
- Protection against common web vulnerabilities.

Reliability:

- We offer 99.9% uptime guarantee.
- Our automated backup technologies keep your data safe.
- Regular system health checks ensure everything runs smoothly.
- And for heavy load, graceful degradation to maintain performance.

Maintainability:

- Modular code structure that is easy to update.

- Comprehensive documentation to guide the way.
- Version control system to track changes with ease.
- Continuous integration to stay in sync.
- Periodic system updates to keep it fresh and secure.

6. Feasibility Analysis

Technical Feasibility:

- We are using .NET Core on the backend.
- Blazor powers our frontend.
- Our database implementations are SQL Server and Azure Blob Storage.
- For AI integration, we're using ML.NET.
- We also communicate with external APIs like the News API and YouTube Data API.

Operational Feasibility:

- A broad audience of general readers, professionals, and students.
- An intuitive user interface to make navigation easier.
- Automated and manual mixed content management for quality control.

Economic Feasibility:

1. **Initial Expenses:**
- Development infrastructure to get us started.
- API subscriptions for basic service access.
- Cloud hosting for scalability.
2. **Income Streams:**
- Premium subscription for premium content.
- Targeted ads for reaching the right audience.
- Affiliate marketing for additional income.

7. Risk Analysis

1. Technical Risks:

- **Integration Problems:** Merging advanced AI features like summarization, sentiment analysis, and personalized recommendations into the mix can be quite difficult. The real challenge is making sure everything plays nicely with the backend of the news website.

- **Scaling Problems:** The site needs to handle millions of users during peak hours without slowing down. Scaling databases, APIs, and AI modules effectively is essential.
- **Database Management:** Handling a large amount of multimedia content—articles, videos, podcasts, and images—on a hybrid database setup (both relational and non-relational) involves careful planning and execution.
- **Cross-Platform Compatibility:** Making the platform compatible and performing optimally on both web and mobile platforms might involve a lot of testing and tweaking.

2. Operational Risks:

- **Team Coordination:** Communication must be sustained and every member must be synchronized with the same goals to achieve project deadlines and manage interdependencies.
- **Skill Deficiencies:** Team members might need to learn new technologies and tools, e.g., advanced AI platforms or database optimization techniques.
- **Resources Availability:** Relying on third-party APIs (e.g., news aggregation and video APIs) and cloud services can hinder development if they crash or change their pricing.
- **Testing and Debugging:** Finding and fixing bugs in complex AI models and getting them to function effectively with the overall system can be challenging.

3. Economic Risks:

- **Budget Overruns:** Unexpected expenses for cloud hosting, API charges, AI software, and multimedia storage can easily exceed the budget.
- **Revenue Streams:** Slippage in deploying monetization models like premium subscriptions, ads, and sponsorships can undermine the financial sustainability of the project.
- **Competition:** Competing with the traditional names like The New York Times and Bloomberg might require extra marketing and special features, which can raise costs

Challenges

Technical Challenges

- **AI Development:** Developing and deploying AI models to offer personalized recommendations, content grouping, and mood detection requires proficiency in machine learning and natural language processing.

- **Multimedia Optimization:** Ensuring the site properly handles high-resolution videos, podcasts, and interactive graphs without performance degradation.
- **Database Design:** Designing a hybrid DB system (relational for regular data and non-relational for multimedia) supporting scalability and durability.
- **Real-Time Notifications:** Building a robust notification system for trending news updates, new content, and subscription events.

Project Management Challenges

- **Timeline Management:** Ensuring regular progress with regulation of dependencies and avoiding delays.
- **Task Distribution:** Allocating tasks to team members to prevent bottlenecks.
- **Testing Across Modules:** Coordinating testing activity across AI, database, frontend, and backend modules to ensure smooth integration.

User Experience Challenges

- **UI/UX Design:** Crafting an easy-to-use and responsive user interface appealing to a broad audience.
- **Content Organization:** Structuring content to enable easy navigation, such as filters, categories, and personalized feeds.
- **Cross-Platform Performance:** Delivering a uniform user experience on web browsers, mobile phones, and tablets.

Workload Distribution

Ananto Nayan Bala:

- Project Planning and Documentation
- AI Integration
- Deployment and Cloud Management
- Backend Development (API)

Shaik Farhan Abid:

- Project Planning and Documentation
- Frontend Development (UI/UX)
- Backend Development (Database)

Md. Sakib:

- Research & Development
- AI Integration

- Testing and Debugging

Atik Ahmed:

- Frontend Development (UI/UX)
- Notification System
- Testing and Debugging

Expanded Workload Details:

Ananto Nayan Bala:

• Responsible for planning project milestones, drafting documentation, and outlining key deliverables.
- Implements AI modules for personalized recommendations and summarization.
- Oversees deployment on cloud platforms and ensures scalability.
- Develops backend APIs for seamless data communication between frontend and database.

Shaik Farhan Abid:

- Designs the frontend interface with a focus on usability and responsiveness.
- Works on backend database setup and integration, ensuring efficient data storage and retrieval.
- Participates in project planning and contributes to defining workflows and timelines.

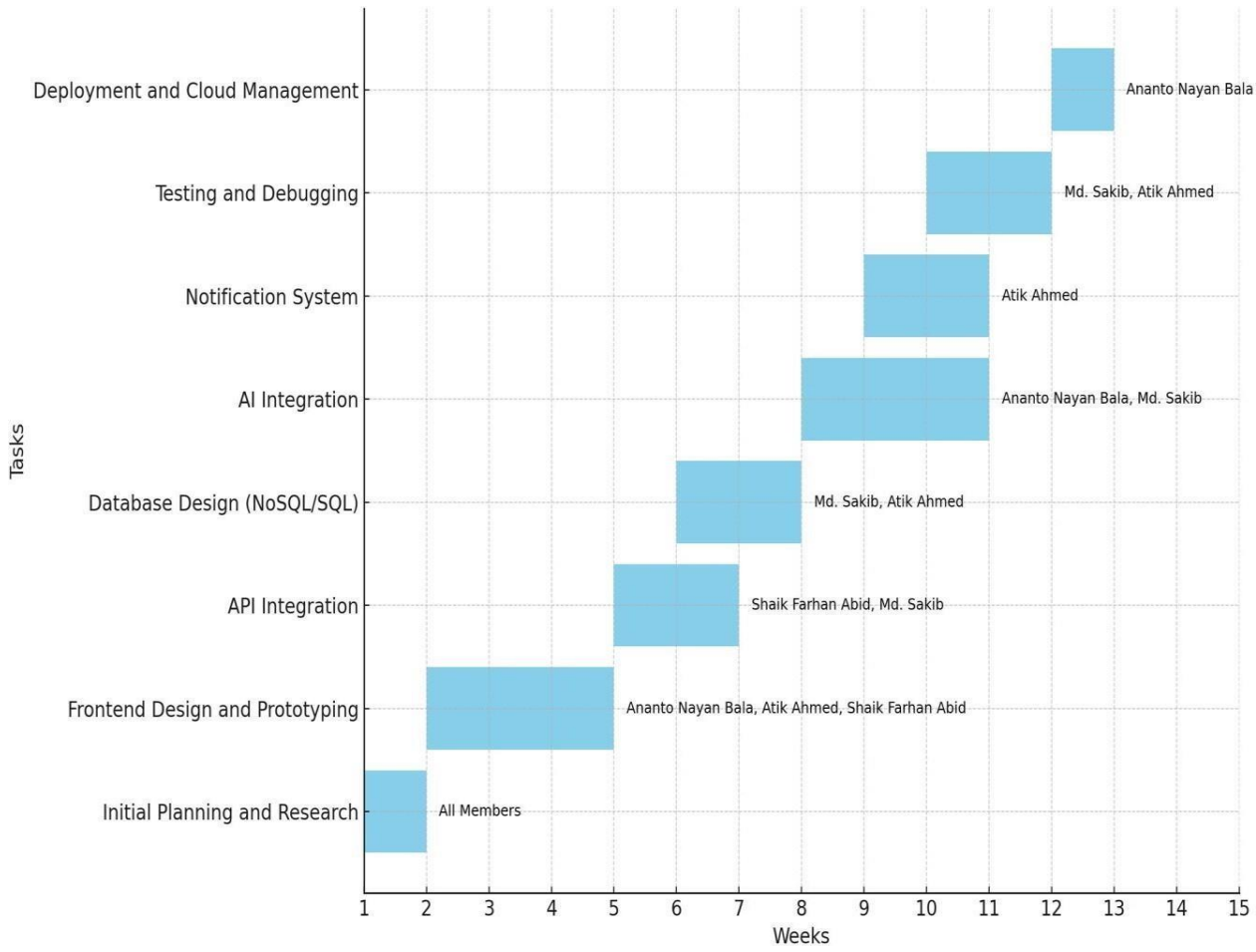
Md. Sakib:

- Conducts research to identify suitable AI frameworks and database tools.
- Collaborates on AI integration, testing models for accuracy and performance.
- Leads testing and debugging efforts across the platform to ensure quality assurance.

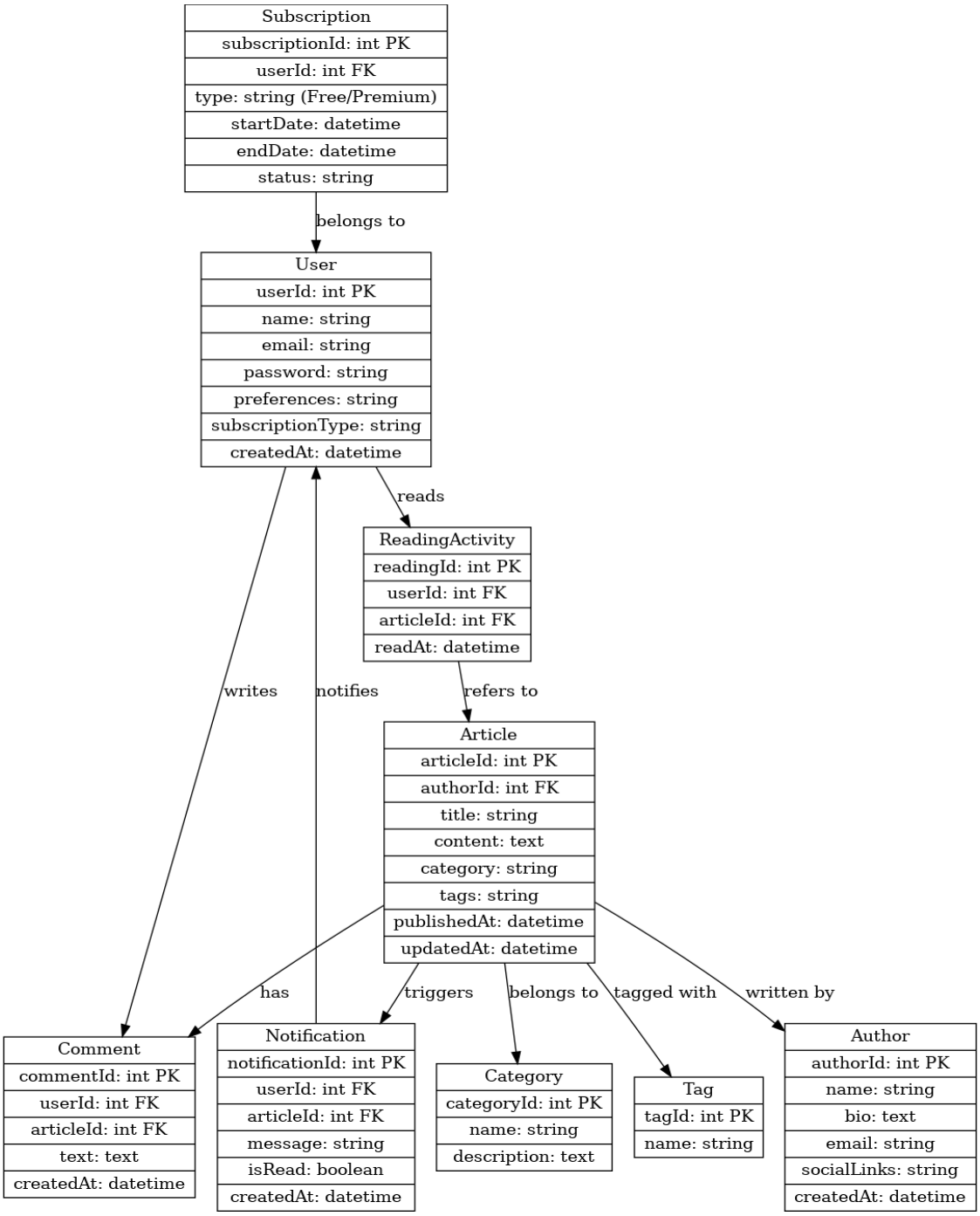
Atik Ahmed:

- Develops the frontend interface, focusing on navigation and interactivity.
- Implements the notification system for user engagement and updates.
- Conducts testing and debugging to resolve UI and functionality issues.

Project Scheduling



Entity Relationship Diagram


Subscription
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userId: int FK
type: string (Free/Premium)
startDate: datetime
endDate: datetime
status: string
belongs to
User
userId: int PK
name: string
email: string
password: string
preferences: string
subscriptionType: string
createdAt: datetime
reads
ReadingActivity
readingId: int PK
userId: int FK
articleId: int FK
readAt: datetime
writes
notifies
refers to
Article
articleId: int PK
authorId: int FK
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tags: string
publishedAt: datetime
updatedAt: datetime
has
triggers
belongs to
tagged with
written by
Comment
commentId: int PK
userId: int FK
articleId: int FK
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createdAt: datetime
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articleId: int FK
message: string
isRead: boolean
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categoryId: int PK
name: string
description: text
Tag
tagId: int PK
name: string
Author
authorId: int PK
name: string
bio: text
email: string
socialLinks: string
createdAt: datetime

Cost-Benefit Analysis

Year	Revenue	Hardware	Cost Software	Cost Utility	Cost Miscellaneous Cost	Total Cost
0	0	120,000	0	4,000	3,000	127,000
1	200,000	80,000	0	2,500	2,000	84,500
2	300,000	40,000	0	2,000	1,500	43,500

Cash Flow: Revenue – Total Cost

Present Value (PV): CF / (1 + r) ^ t

Net Present Value (NPV):∑ PV

Discount Factor (r): 20%= 0.2

Year	Cash Flow	PV
0	–127,000	–127,000
1	115,500	96,250
2	256,500	178,125

NPV = 147,375

Here, the Net Present Value (NPV) is positive, indicating tangible benefits. However, the project offers several **tangible benefits**, such as:

- Enhancing AI integration and web development skills.
- Adding valuable experience to team members' CVs.
- Providing practical experience in teamwork.
- Simulating real-life workflows.
- Contributing to academic progress.

Use Case Diagram for Parody Journal

Actors

1. Admin

Description: The administrator responsible for overseeing the platform.

Role: Manage users, approve content, and ensure proper functioning of the system.

2. News API
Description: An external service that provides news articles for various sources. Role: Fetch and update articles automatically.

3. Reader

Description: End-user consuming content on the platform. Role: Browse, search, and engage with the news and articles.

4. AI

Description: An intelligent system that enhances user experience.

Role: Provide personalized content recommendations and summarize articles.

Use Cases

1. Login Actors: Admin

Description: Users log in to access the system's features. Priority: High.

2. Manage Writers Actors: Admin.

Description: Admin oversees and manages content writers. Priority: Medium.

3. Approved Articles Actors: Admin

Description: Admin approves articles for publication; the server processes the updates. Priority: High.

4. Manage Articles Actors: Admin.

Description: Admin can update, delete, or modify articles. Priority: Medium.

5. Update Articles Actors: News API.

Description: Server handles real-time updates to content. Priority: High.

6. Browse Articles Actors: Reader.

Description: Readers explore available articles. Priority: High.

7. Search Articles Actors: Reader.

Description: Readers search for specific articles using keywords or filters.

Priority: High.

Detailed Descriptions of Different Scenarios

1: Login

Prerequisite: User must have an account.

Steps:

User navigates to the login page.

User enters credentials (email and password). System validates the credentials.

User is redirected to their dashboard.

2: Browse Articles

Prerequisite: Reader must access the platform.

Steps:

Reader navigates to the homepage.

System displays available articles, including trending or recommended ones. Reader clicks on an article to view the content.

3: Fetch Articles

Prerequisite: The system must be connected to the News API.

Steps:

The system sends a request to the News API.

The News API retrieves the latest articles from various sources. The system processes and categorizes the fetched articles.

4: Personalized Content

Prerequisite: Reader must access the platform. Steps:

AI analyzes reader preferences, browsing history, and trending topics. AI displays a personalized feed with suggested articles.
Text BoxReader selects and views articles from the recommendations.

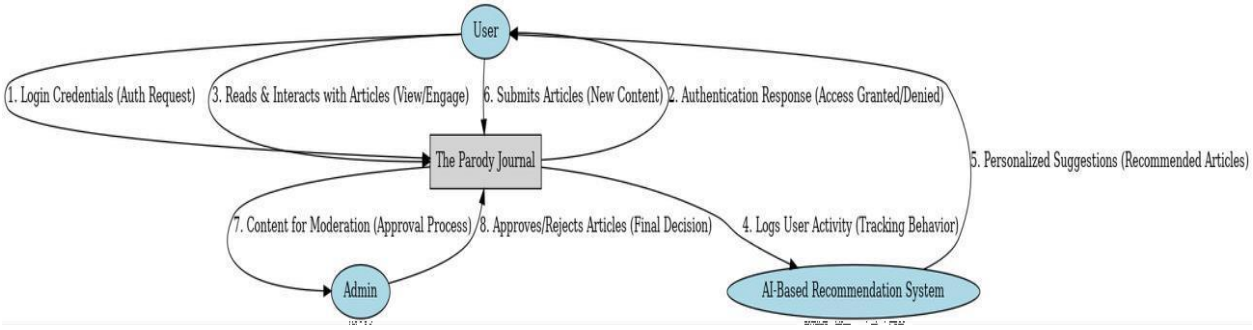
Data Flow Diagram (DFD) for The Parody Journal

1. **Main Process & Sub-Processes**
Main Process: The Parody Journal A satirical news portal where users can read, interact, and submit parody articles.

Sub-Processes:

- **User Authentication** – Login & registration.
- **Browsing & Reading Satirical News** – Users explore categories.
- **User Interaction** – Likes, comments, and sharing.
- **AI-Based Recommendations** – Suggests articles based on engagement.
- **Content Submission** – Users submit their own parody stories.
- **Admin Moderation** – Admin reviews submissions and manages users.

Context Level Diagram
The Context Diagram represents external entities interacting with The Parody Journal system.



Entities:

- **User** – Reads, interacts, and submits articles.
- **Admin** – Moderates content and users.
- **AI-Based Recommendation System** – Analyzes user activity.

Data Flow:

- User → Login Credentials → The Parody Journal
- The Parody Journal → Authentication → User Database
- User → Reads & Interacts with Articles → News System
- News System → Logs User Activity → AI System
- AI System → Generates Personalized Suggestions → User
- User → Submits Articles → Content Review System
- Admin → Approves/Rejects Articles → Published News

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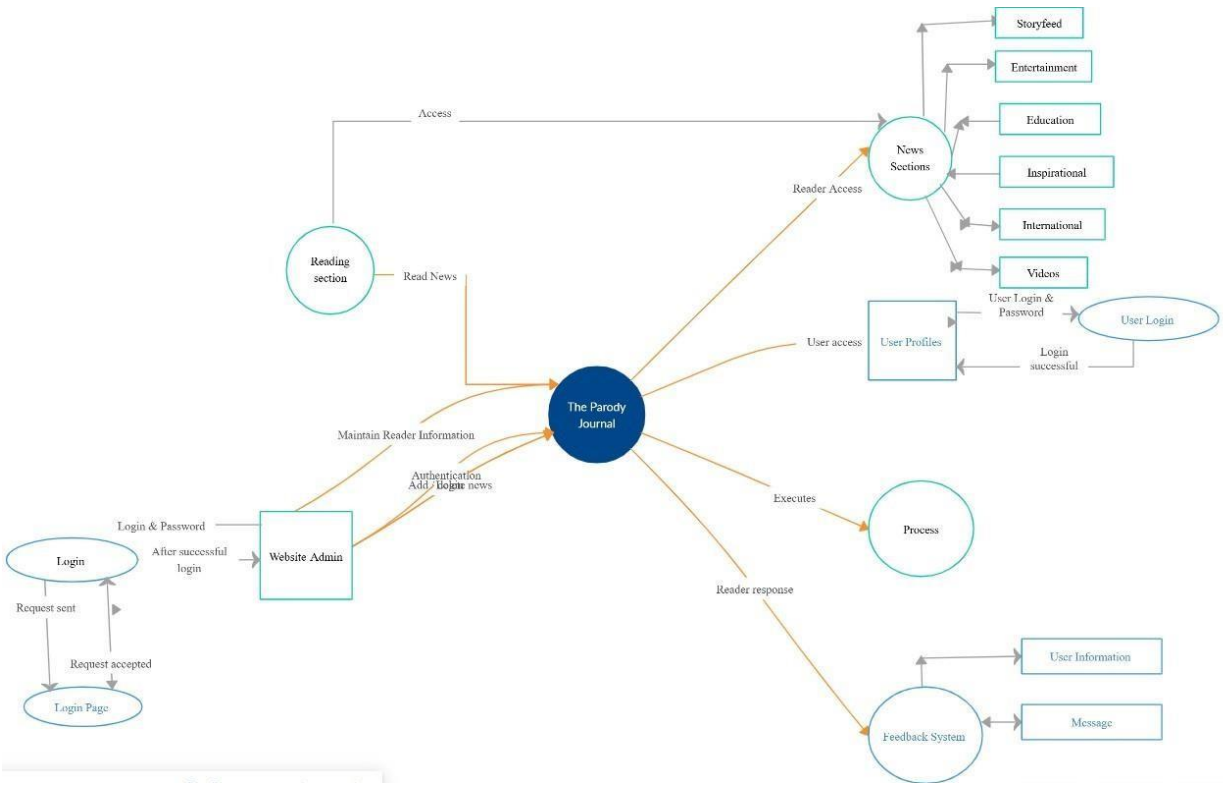
2. **Level 0 Diagram**



Processes & Data Flow:

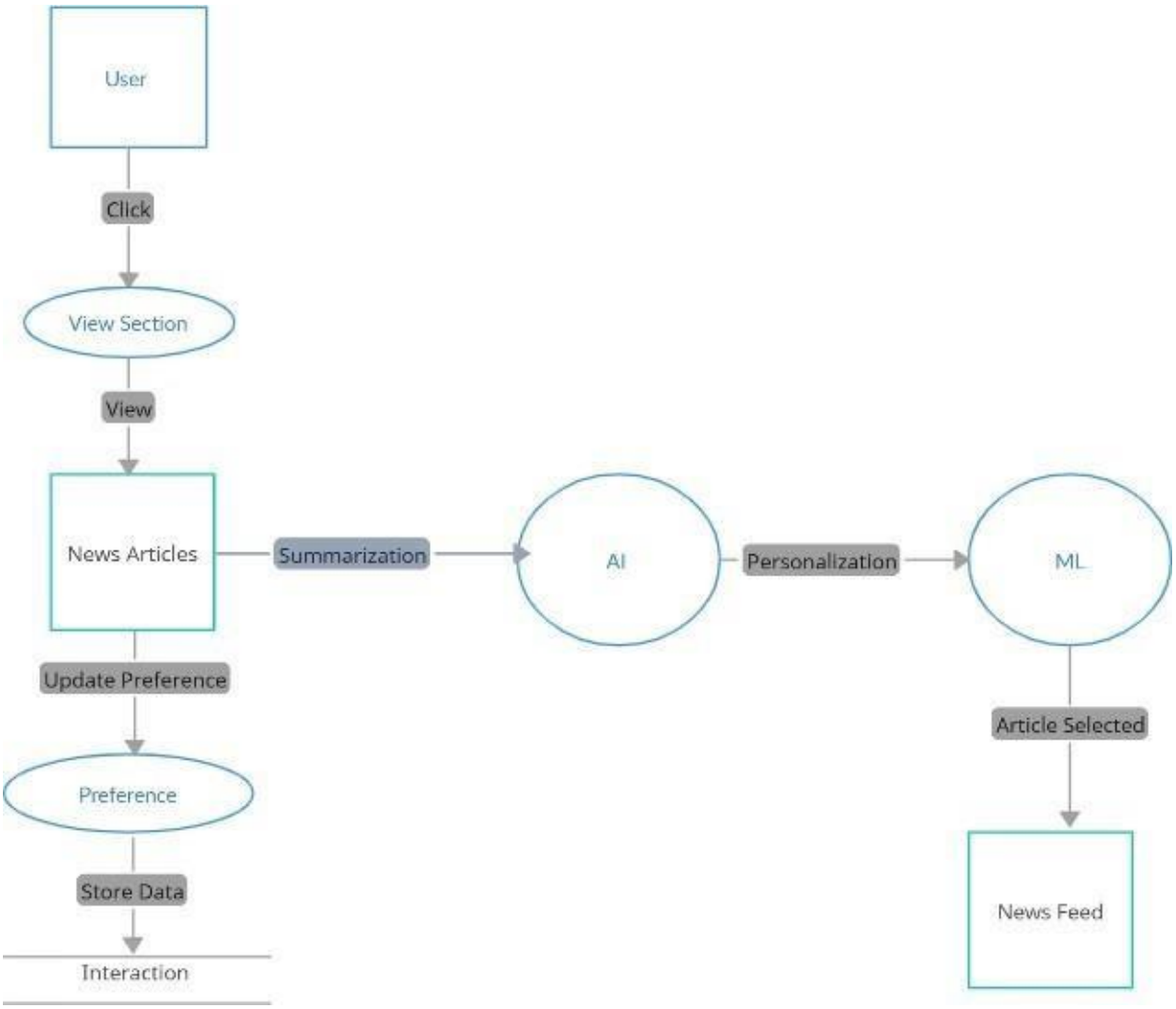
- **User Login & Authentication**
  - Users log in → System verifies credentials → Access granted.

News Browsing & Reading
  - Users select categories → Articles are retrieved.

User Interactions (Likes, Comments, Shares)
  - Users engage with content → Data is updated.

Admin Management
  - Admin moderates content & users → Updates system.

Level 1 Diagram (Detailed View of Process)



News Browsing & Reading

- **Input:** User request for articles
- **Process:** Fetch news from database
- **Output:** Display satirical news

User Interactions (Likes, Comments, Shares)

- **Input:** User engagement actions
- **Process:** Update interaction data
- **Output:** Store engagement data in the AI system

AI-Based Recommendations

- **Input:** User browsing & interaction history
- **Process:** Generate personalized content suggestions
- **Output:** Display recommended articles

AI-Based Text Summarization

- **Input:** Raw news articles
- **Process:** Extract key points and generate concise summaries
- **Output:** Display summarized articles for quick reading

Project Contributions:

Shaik Farhan Abid

- Designed the Login Page
- Developed the Homepage
- Designed & Navigated the Text Summarization feature with Front End

Atik Ahmed

- Implemented the Sign-in Page
- Developed the Forgot Password Page
- Integrated the News API

Ananto Nayan Bala

- Built the Recommendation System, including:
  - Article Recommendation using Machine Learning
  - Tracking & Updating/Removing Preferences on User Click
  - Category-Wise Article Distribution for ML Training

Md. Sakib

- Developed the Text Summarization Feature
- Integrated the summarization system with the Postman Agent

Conclusion:

Parody Journal's implementation has successfully addressed the growing demand for an interactive, personalized, and AI-driven news website. With the inclusion of diverse content, multimedia support, and AI-driven recommendations, the website offers an interactive and personalized reading experience to the users. User engagement is ensured by social and interactive features, while scalability, performance, and security are ensured with the scalable backend implemented using .NET Core and Blazor. Through extensive research and studies of viability, not just have we technologically

established Parody Journal robust but also environmentally adoptable by the users and market-sustainable. AI-based sentiment analysis, summarization, and recommendation algorithms being used differentiate this site from generic news websites. Also, the model of monetizing through premium subscription, targeted advertising, and affiliate marketing makes it sustainably viable. In the coming years, continuous growth, feature improvements, and AI updates will be key to keeping up with evolving user demand and market trends. With its new reading news model in a read format, Parody Journal will change the way consumers interact with information and news on the internet.